

AI and the future of energy

Posted by u/cheinyeanlim • 0 points • 0 comments

The expansion of AI is set to significantly increase electricity demand in the US.

According to a Wells Fargo analysis, demand could rise 20% by 2030, driven by growth in AI, semiconductor production, battery manufacturing, and the electrification of vehicles.

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Data centres catering to AI alone are projected to require an extra 323 terawatt hours—vastly more than the current annual consumption of New York City, which stands at 48 terawatt hours.

****Here's what you should know:****

- * Technological advancements, particularly in AI, are expected to increase US electricity demand significantly by 2030.
- * As a quick-response energy source, natural gas could be critical for bridging gaps when renewable sources fall short.
- * The Southeast US is crucial, with major growth in data centres and energy demand anticipated, supported by strategic infrastructure projects like pipelines.

This demand increase presents challenges for major tech companies like Amazon, Google, Microsoft, and Meta, which are committed to using renewable energy sources.

However, the variability of solar and wind power makes these sources unreliable for consistent high-volume demand.

Natural gas, capable of rapidly meeting large energy needs, is seen as an essential interim solution.

It is expected to cover 60% of the power demand increase caused by AI and data centres.

Energy companies there are preparing for significant changes, with Dominion Energy predicting that demand in Northern Virginia will double by 2030.

EQT Corp, a leading gas producer, considers this region strategic for growth, aided by facilities like the Mountain Valley Pipeline.

****But, AI can speed up renewables****

But it's not all doom and gloom. AI could solve the very issue it is creating. It can make the energy sector more efficient and better at integrating renewable sources.

It helps predict energy needs, streamline grid operations, and manage energy resources effectively.

These advancements are key for maintaining reliable energy as we face climate challenges and variable demand.

The Southeast US is crucial for upcoming energy projects, supported by infrastructure like the Mountain Valley Pipeline.

AI's ability to improve efficiency and integration of renewables is vital for a sustainable energy future.

So, use more energy - to learn how we can use less.

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